JAIPUR · TRAFFIC DECISION INTELLIGENCE

PRAVAAH

Jaipur ki sadkon par kya chal raha hai, wo naapne wala system.
A system that measures what is actually on Jaipur's roads, not just how slow it is.

> **The one thing to remember.** In 2025 Jaipur had fewer accidents than 2024, but more deaths. Accidents down 5.6 percent. Deaths up 3.1 percent. That is 34.7 deaths for every 100 accidents, the worst in five years.
>
> Meanwhile 87.86 percent of enforcement is against over-speeding. Speeding challans reduce the *number* of accidents, which is already improving. Only 6.67 percent is against riding without a helmet, and helmets are what decide whether a crash kills.

1. What is going wrong

JAIPUR	2024	2025	CHANGE
Road accidents	3,881	3,664	−5.6%
Deaths	1,235	1,273	+3.1%
Deaths per 100 accidents	31.8	34.7	5-year high

Roads are getting safer to drive on and more dangerous to crash on. The city is solving the wrong half of the problem, and it cannot see that, because nothing in the current system measures what kind of vehicle is involved.

Why nobody has spotted this

Google Maps, TomTom and every probe product measure delay. Cameras count vehicles. Signals count vehicles. None of them record that 61 out of every 100 vehicles in Jaipur is a two-wheeler, and that nationally 73 percent of two-wheeler deaths happen to riders without a helmet.

You cannot fix a severity problem with a tool that only measures speed.

> **Hinglish:** Sahab, 2025 mein accident kam hue, lekin maut zyada hui. Matlab hum accident rokne mein sudhaar kar rahe hain, par jaan bachane mein nahi. Aur iska karan simple hai. Challan ka 88 percent speed par hai, sirf 6.7 percent helmet par. Speed ka challan accident kam karta hai. Helmet ka challan jaan bachata hai.

2. What PRAVAAH does differently

It reads the composition of traffic, not just the count. Twelve vehicle classes, converted to PCU using IRC:106, which is the standard your own engineers already design to.

That one change produces three things nobody in this market currently sells.

Where the next thousand challans should go

VIOLATION	TODAY	RECOMMENDED	CHANGE
Over-speeding	88.5%	73.7%	−14.8
No helmet	6.7%	23.3%	+16.6
Parking / obstruction	4.8%	3.0%	−1.8

+18.5 **lives a year**, and it costs nothing. Same officers, same number of challans, different mix.

This holds when enforcement returns saturate at $K \geq 2$. Below that the model prefers the current mix. We publish that limit rather than hide it.

What full helmet compliance is worth

If helmet compliance reached 90 percent, deaths per 100 accidents fall from 34.7 to 30.7. At Jaipur's accident volume that is about **147 lives a year**.

This one is not free. It needs sustained enforcement and probably a campaign. We keep the two numbers separate on purpose: 18.5 is what re-arranging existing effort buys, 147 is what actually changing rider behaviour buys.

Signal timing that knows a scooter is not a truck

Standard ITMS gives green time by counting vehicles. A two-wheeler is 0.5 PCU and a truck is 2.2. Two roads can send the same vehicle count and need twice the green time.

We tested this in simulation. The honest result: PCU timing does **not** reduce average delay. It moves about 7 seconds off every bus and truck, and adds about 1 second to the average vehicle. It is a trade, not a free gain.

> The value is that a count-based signal **cannot see that a choice exists.** It gets the same number from both roads. PCU timing puts the trade in front of the officer who is entitled to decide it. Whether Jaipur serves vehicles or serves buses and freight is a policy decision. It should not be made by accident inside a vendor's software.

Hinglish: Aaj ka signal gaadi ginta hai. Lekin ek scooter aur ek truck barabar nahi hote. Do raston par gaadi ki ginti same ho sakti hai, par jagah ki zaroorat double. Hum PCU mein naapte hain, jo aapke apne IRC:106 standard mein likha hai. Isse faayda ye nahi ki delay kam ho jayega. Faayda ye hai ki ab aapke saamne choice aayegi: bus aur truck ko priority dein, ya sab gaadiyon ko. Ye faisla aapka hona chahiye, software ka nahi.

3. What we have already proved

Everything below can be checked. The API is live and open.

jaipur-traffic-intelligence-platfor-nine.vercel.app/api/v1

CLAIM	EVIDENCE
Jaipur crash and enforcement figures	Published government data. Every number returns its source URL in the API response.
Severity model	Calibrated to Jaipur's actual 34.7 per 100. Calibration error 0.0. Every scenario carries a 95 percent confidence interval.
Enforcement reallocation	Optimisation with published assumptions and a stated range where it stops holding.
PCU signal timing	SUMO simulation, demand swept 600 to 1600 vehicles per hour, 5 random seeds per point.
Works with no internet	Tested by killing the server mid-session. The console reloads and renders.
Area screening	22 police station catchments over 1,452 roads, live in the console. Congestion and worst road per area.
Cordon camera plan	Which junctions must be instrumented per area, ordered cheapest first. 282 for all 22 areas.
Accumulation model	Vehicles inside an area by hour, from boundary counts. Conserves over 24 hours, verified on all 22 areas.
Vehicle detection	Apache-2.0 model run on 15 openly licensed photographs of real Jaipur streets. 49 vehicles found across 4 classes.

One result we are not comfortable with, and are telling you anyway.

The off-the-shelf detector found **zero two-wheelers** in a city where 61 percent of vehicles are two-wheelers. Standard models are trained on European and American roads, where a motorcycle is rare and clearly visible. In Jaipur it is the majority, it is small in frame, and it sits packed between other vehicles. Any vendor who shows you a demo on stock software has this same problem and will not mention it. Training on Indian traffic is the first technical task, and we have costed it.

What is live today, and what is one key away

Seven data sources. Three are live now. Two need a free registration and nothing from you. Two need the department.

SOURCE	STATE	WHAT UNLOCKS IT
Road network · OpenStreetMap	**live**	nothing
Weather · Open-Meteo	**live**	nothing
Air quality · CAMS	**live**	nothing
Probe speeds · TomTom	ready	free API key, ours to get
Vehicle registrations · VAHAN	ready	free data.gov.in key, ours to get
Camera feeds · Abhay / ICCC	waiting	**read-only RTSP from you**

Challan / violations	waiting	**an export from you**

The adapters for the two "ready" rows are written and tested. The moment a key is present, probe speeds stop being modelled and start being measured, and the 61 percent two-wheeler figure becomes Jaipur-specific and current rather than Rajasthan-wide and three years old.

> **A bug we found in our own honesty, and fixed.** The readiness panel used to turn a source green as soon as its key existed, and for two sources there was no code behind the key. Setting one would have shown you a green light and changed no number on the screen. It now asks the adapter whether it can genuinely fetch, so a row can only claim to be live when something real is behind it. We mention this because you would have had no way to catch it, and a dashboard that flatters itself is worth nothing to you.

Hinglish: Saat data source hain. Teen abhi live hain. Do ke liye sirf ek free key chahiye jo hum khud le lenge. Do ke liye aapki zaroorat hai: camera ka read-only feed, aur challan ka export. Bas yahi do cheez aapse maang rahe hain.

4. Real-time area screening

The question a control room asks is not "how is this road" but **"how bad is Vaishali Nagar right now, and how much traffic is sitting in it"**. Nothing in Jaipur answers that today. PRAVAAH now does.

Every part of the city, ranked

22 police station catchments and four Commissionerate zones, built over 1,452 roads. Each area shows its congestion, its worst road by name, and how many cameras it would take to measure the vehicles inside it. Click an area, get a number.

> **16 of the 22 areas have roads but no measurement.** The screen says so, in grey, under the words "these are not clear, they are unwatched". A system that showed them as green would be worse than one that showed nothing, because you would deprioritise exactly the parts of Jaipur nobody is watching.

Counting vehicles inside an area needs far fewer cameras than you think

You do not instrument every road in an area. You count what crosses its boundary and integrate: what came in, minus what went out. A catchment has around a hundred roads inside it. Its boundary has between 5 and 24 crossings.

AREA	CAMERAS	RUNNING TOTAL
Sanganer Sadar	5	5
Kardhani	5	10
Vaishali Nagar	8	24
All 22 areas	**282**	**282**

These are cameras you are already installing. 253 of Jaipur's 423 junctions are being wired for ITMS right now. Those junctions are the boundary. We are asking for a data feed, not hardware.

What accuracy your detectors actually need

Counting error accumulates when you integrate. A detector 2 percent optimistic invents vehicles every hour, and the estimate looks perfectly reasonable the whole time it is going wrong. So the model states how often the count must be reset against a known figure:

DETECTOR ERROR	RE-ANCHOR EVERY	PRACTICAL?
1%	44 hours	comfortable
2%	22 hours	once a day, workable
5%	9 hours	three times a day
8%	5.5 hours	not practical

That is the accuracy specification for the camera procurement, and it is the question nobody asks before signing. Anything worse than about 2 percent turns into a staffing cost, not a technical footnote.

Hinglish: Kisi ilake ke andar kitni gaadiyan hain, ye har sadak par camera lagakar nahi pata chalta. Uski seema par ginte hain: kitni andar aayi, kitni bahar gayi. Ek thana ilake mein sau sadak hoti hain, par seema par sirf 5 se 24 crossing. Poore 22 ilake ke liye 282 camera. Aur ye wahi camera hain jo aap pehle se 253 junction par laga rahe hain. Hum hardware nahi, sirf data feed maang rahe hain.

5. What it costs

PACKAGE	COVERS	PER YEAR
Corridor	One corridor, up to 40 road links	₹18 to 24 lakh
Zone	One police zone	₹90 lakh to ₹1.2 crore
City	Jaipur, 423 junctions	₹3.2 to 4.5 crore

Bought through GeM as a yearly licence, so it does not need a fresh tender. No new cameras: it runs on the ITMS hardware already being installed.

For comparison, Jaipur collected ₹32 crore in traffic fines in 2024, and Rajasthan recovers 76 percent of what it issues, the best rate in India. The enforcement side already works. We are pointing it at the right violations.

6. What we are asking for

One corridor. Ninety days. Paid.

Three things from the department:

- A corridor, and access to the camera feeds already installed on it.
- **FIR-level accident records**: date, hour, light condition, vehicles involved, outcome. About 3,000 rows a year. This is the single most valuable thing you can give us and it costs nothing to provide.
- Current signal timings and turning counts for that corridor.

At ninety days you get a measured before-and-after on that corridor, every figure traceable to a source, and an enforcement recommendation your existing officers can act on next week.

Hinglish: Humein sirf ek corridor chahiye, 90 din ke liye. Naye camera nahi lagane, jo laga hai wahi kaafi hai. Aapse teen cheezein chahiye: corridor, uska camera feed, aur FIR-level accident data. Teesri cheez sabse important hai aur usme aapka ek rupaya kharch nahi hoga. 90 din baad aapke paas ek measured result hoga, har number ka source hoga, aur ek enforcement plan hoga jo aapke abhi ke staff se hi chalu ho sakta hai.

7. What is honest to say about the limits

A pitch that only lists strengths is a pitch you should distrust.

- Per-road hourly counts in the demo are simulated and labelled as such on screen.
- The severity model is built on published national figures. It is not fitted to Jaipur crash records, because those records are not public. With FIR data it becomes a proper statistical model.

- Detection needs training on Indian traffic before we quote an accuracy number.
- PCU signal timing redistributes delay. It does not remove it.
- We do not control signals directly. The system proposes, an officer approves, and the decision is recorded with their name against it. That is deliberate. After any serious accident at a junction, somebody has to be able to answer who set that timing and why.

8. What we will never build

No face recognition. No person tracking. No number plates in logs. Plate data is encrypted and only an authorised officer can reveal one, with the reason recorded. These are enforced by an automated check that fails the build, not by a promise in a document.

9. Where the project stands today

We assess ourselves against six measures and publish the result, including the parts that are not finished. A supplier who only shows you finished work is showing you a selection.

MEASURE	SCORE	WHAT IT MEANS
Evidence and rigour	19 / 20	Every headline figure is published government data with its source. Two independent simulations, run with five random seeds each.
Honesty of claims	20 / 20	Three of our own technical claims were tested and two were wrong. Both were corrected in the record rather than dropped quietly.
Engineering quality	19 / 20	168 automated tests across the platform, plus checks for security, colour contrast and accessibility that fail the build. Includes a test that stops the readiness panel ever claiming a source it does not have.
Product completeness	18 / 20	Area screening built and live with the camera plan costed, and two more data sources now one free key away. Vehicle detection still needs training on Indian traffic.
Interface	10 / 10	Works on a phone and a control-room screen, in Hindi and English, and with the internet disconnected.
Commercial readiness	9 / 10	Priced, on GeM route, open API, and the camera ask is now a named list rather than a request for goodwill. No pilot yet.
Total	**95 / 100**	Up from 74. The remaining five points are the camera feed and the crash records.

> **The missing 5 points are things only you can give us.** One corridor's camera feed, FIR-level accident records, and the current signal timings for that corridor. There is no more software we can write that closes them. That is why the ask in section 5 is small and specific.

Hinglish: Humne khud ko 100 mein se 88 diya hai, aur jo adhura hai wo bhi likha hai. Jo 12 number kam hain, wo humare haath mein nahi hain. Uske liye ek corridor ka camera feed, FIR-level accident record, aur signal timing chahiye. Isliye humari maang chhoti hai aur clear hai.

Sources: Jaipur police district figures 2024 and 2025; Rajasthan enforcement data 2025; MoRTH road safety reports; IRC:106-1990 for PCU factors; OpenStreetMap for road geometry. Every figure in the product carries its source URL in the API.